Software Engineering - Lab 1

Name : Nishkaa V

Topic : Community Lost & Found Matching Network

SRN : PES2UG24CS907

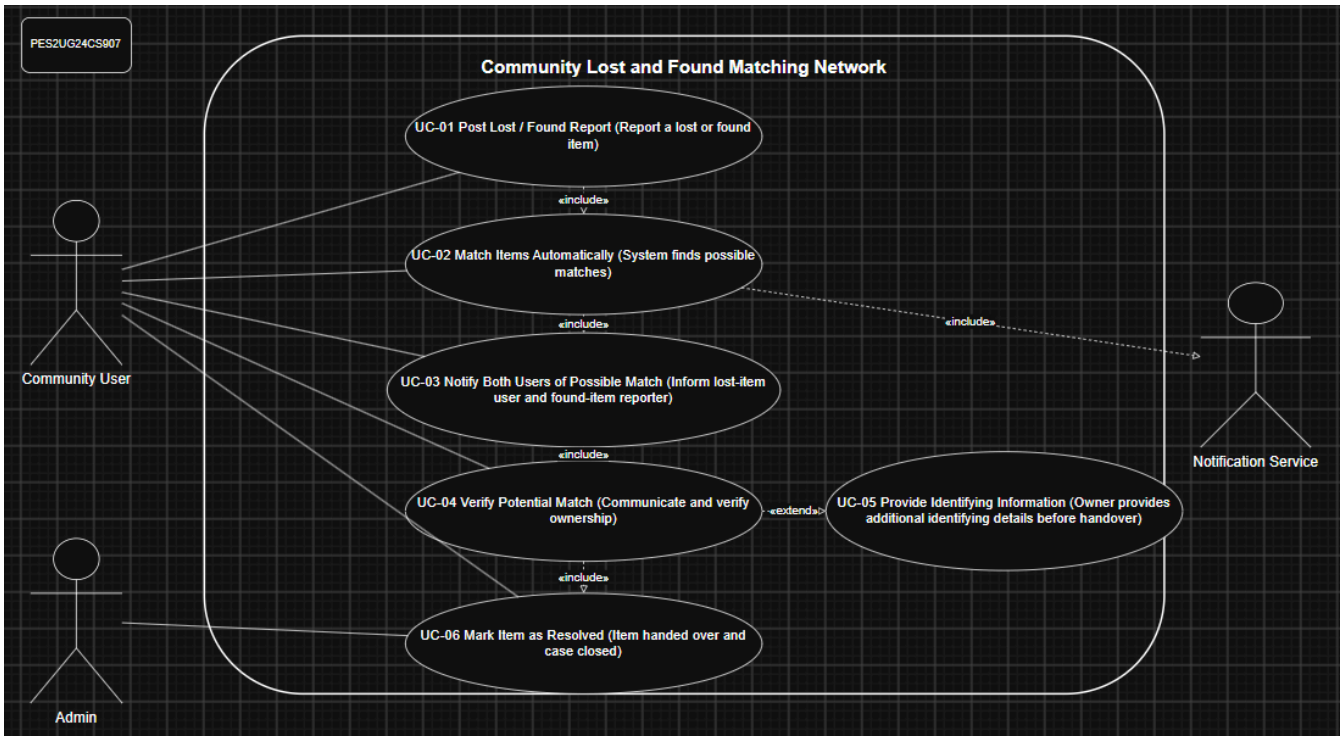
Class : 5 - B

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow a user to report a lost item by entering the item category, description, approximate location, date, and photographs.	High	A report is accepted only when category, description, approximate location, and date are provided; photographs may be attached. The saved report displays the submitted details.	Core function explicitly required for reporting lost items.	Check that all required lost-item fields are captured.
FR-002	Functional	The system shall allow a person who finds an item to register a found-item report with the item's relevant details.	High	A found-item report can be submitted and saved with the entered item details; the system assigns the report an identifiable record.	Required so found items can be compared with lost-item reports.	Specify the minimum details expected for a found item.
FR-003	Functional	The system shall automatically search for possible matches between lost and found reports using category, description, location, date, and keywords.	High	For a submitted report, the system performs a match search using all five specified attributes and produces a result indicating whether possible matches were identified.	The automatic matching capability is the central purpose of the platform.	Ensure the matching logic uses all five scenario attributes.
FR-004	Functional	The system shall notify both users when a possible match is identified between a lost-item report and a found-item report.	High	When a possible match is identified, a notification is generated for both the user who reported the lost item and the person who reported the found item.	The scenario explicitly requires both users to be notified of possible matches.	Verify that both sides of the match receive the notification.
FR-005	Functional	The system shall allow users to communicate	High	For a possible match, the involved users can	Supports verification and the required	Ensure identifying

		through the platform to verify a possible match and shall allow the owner to provide additional identifying information before handover.		exchange messages through the platform, and the owner can submit additional identifying information before the item is handed over.	pre-handover identification step.	information is requested before handover
NFR-001	Nonfunctional	The system shall complete the automatic possible-match search for a submitted report within 5 seconds under normal operating conditions.	Medium	In a test with 100 consecutive representative report searches under normal operating conditions, at least 95 searches complete within 5 seconds.	Provides a measurable responsiveness target for the platform's central automatic matching function; this target is a lab-defined constraint because the scenario gives no timing value.	Validate the 5-second threshold using a timed test.
NFR-002	Nonfunctional	The system shall restrict report and communication access so that a user can access only their own reports and conversations involving their possible matches, while administrators can access reports for investigation.	Medium	In access-control tests, a user cannot open another user's unrelated report or conversation, while an administrator can open a report submitted for investigation.	Protects lost/found reports and identifying information; this is a lab-defined security constraint because the scenario does not specify an access-control policy.	Test both permitted and denied access cases.

UML - Use Case Diagram



Use Case-04: Verify Potential Match

Use Case ID	UC-04
Use Case Name	Verify Potential Match
Primary Actors	Lost Item Reporter; Found Item Reporter
Supporting Actor	Administrator (only when a suspicious/fraudulent claim is reported)
Trigger	The system identifies a possible match between a lost-item report and a found-item report.
Preconditions	A lost-item report and a found-item report have been registered; the system has identified a possible match.
Postconditions	If ownership is verified and the item is returned, the relevant report is marked as resolved. If verification fails, the report remains unresolved.

Main Success Scenario

1. The system identifies a possible match between a lost-item report and a found-item report using category, description, approximate location, date, and keywords.

2. The system sends a notification to the user who reported the lost item and to the person who reported the found item.

3. Both users open the possible-match notification and communicate through the platform.

4. The users exchange relevant information to verify whether the found item belongs to the person who reported it lost.
5. If required, the owner provides additional identifying information before the item is handed over.

6. The users verify that the item belongs to the owner.

7. The item is handed over to the owner.

8. The system marks the relevant report as resolved.

9. The use case ends successfully.

Alternate Flow – Ownership Cannot Be Verified

5a. The owner cannot provide sufficient additional identifying information when requested.
5a1. The users do not verify that the found item belongs to the owner.
5a2. The item is not handed over.
5a3. The report remains unresolved and no successful return is recorded.